Dynamic Array

HackerRank

- Declare a 2-dimensional array, arr, with n empty arrays, all zero-indexed.
- Declare an integer, $lastAnswer$, and initialize it to 0.

You need to process two types of queries:

1. Query: $1\ x\ y$
   - Compute $idx = (x \oplus lastAnswer)$.
   - Append the integer y to $arr[idx]$.
2. Query: $2\ x\ y$
   - Compute $idx = (x \oplus lastAnswer)$.
   - Set $lastAnswer = arr[idx][y \% size(arr[idx])]$.
   - Store the new value of $lastAnswer$ in an answers array.

Notes:
- $\oplus$ is the *bitwise XOR* operation, which corresponds to the `^` operator in most languages. Learn more about it on Wikipedia.
- $\%$ is the modulo operator.
- Finally, $size(arr[idx])$ is the number of elements in $arr[idx]$.

Function Description

Complete the $dynamicArray$ function with the following parameters:
- $int\ n$: the number of empty arrays to initialize in arr
- $int\ queries[q][3]$: 2-D array of integers

Returns

- $int[]$: the results of each type 2 query in the order they are presented

Input Format

The first line contains two space-separated integers, n, the size of arr to create, and q, the number of queries, respectively.
Each of the q subsequent lines contains a query string, $queries[i]$.

Constraints

- $1 \leq n, q \leq 10^5$
- $0 \leq x, y \leq 10^9$
- It is guaranteed that query type 2 will never query an empty array or index.

Sample Input

```
STDIN    Function
-----    --------
2 5      size of arr[] n = 2, size of queries[] q = 5
1 0 5    queries = [[1,0,5],[1,1,7],[1,0,3],[2,1,0],[2,1,1]]
1 1 7
1 0 3
2 1 0
2 1 1
```

Sample Output

```
7
3
```

Explanation

Initial Values:
$n = 2$
$lastAnswer = 0$
$arr[0]$ = [ ]
$arr[1]$ = [ ]

Query 0: Append 5 to $arr[(\ (0 \oplus 0)\ \%\ 2\ )] = arr[0]$.
$lastAnswer = 0$
$arr[0]$ = [5]
$arr[1]$ = [ ]

Query 1: Append 7 to $arr[(\ (1 \oplus 0)\ \%\ 2\ )] = arr[1]$.
$arr[0]$ = [5]
$arr[1]$ = [7]

Query 2: Append 3 to $arr[(\ (0 \oplus 0)\ \%\ 2\ )] = arr[0]$.
$lastAnswer = 0$
$arr[0]$ = [5, 3]
$arr[1]$ = [7]

Query 3: Assign the value at index 0 of $arr[(\ (1 \oplus 0)\ \%\ 2\ )] = arr[1]$ to $lastAnswer$. Store $lastAnswer$ in your answer array. $lastAnswer = 7$
$arr[0]$ = [5, 3]
$arr[1]$ = [7]

Query 4: Assign the value at index 1 of $arr[(\ (1 \oplus 7)\ \%\ 2\ )] = arr[0]$ to $lastAnswer$. Store $lastAnswer$ in your answer array. $lastAnswer = 3$
$arr[0]$ = [5, 3]
$arr[1]$ = [7]

Return your answer array [7, 3]. The code stub prints its elements on separate lines.